

Rithish S

Software Development Engineer Intern | Developer Tools, AI Agents (MCP/RAG) & Scalable Systems

+91 99522 52304 | rithishcodespace@gmail.com | LinkedIn | GitHub | LeetCode

EDUCATION

Bannari Amman Institute of Technology

Bachelor of Engineering in Computer Science and Engineering

CGPA: 8.28/10

Sathyamangalam, Tamil Nadu

Aug. 2024 – May 2028

TECHNICAL SKILLS

AI, Agents & RAG: LLMs, RAG, MCP, AI Agents, Vector Search (Pinecone), Local LLMs (Qwen2.5, Ollama), Embeddings, Prompt Engineering

Languages & Core CS: Java, C++, Python, JavaScript (ES6+), TypeScript, SQL, OOP, DBMS, Operating Systems, Computer Networks

Backend & Systems: Node.js, Fastify, Express.js, REST APIs, Redis (Distributed Queues & Locks), BullMQ, Clean Architecture

Databases & Cloud: PostgreSQL, MySQL, Redis, AWS (Cloud Practitioner), Oracle Cloud (OCI Certified), Docker, Kubernetes, Linux (POSIX)

Developer Tools & QA: Git, GitHub Actions, AST Static Analysis, ESLint, Prisma ORM, Automated Testing (120+ Tests)

OPEN SOURCE & DEVELOPER TOOLS

ESLint Core | JavaScript, AST, Static Analysis, IEEE 754 Floating-Point Precision [PR #21218](#)

- Contributed a merged production fix to ESLint's **no-loss-of-precision** static analysis rule, correctly distinguishing literal numeric zeros from non-zero decimals that underflow under IEEE 754 binary64 specifications.
- Authored regression test suites covering subnormal values, signed zero edge cases, and non-decimal numeric literals, passing **43 automated CI verification checks** prior to core merge into the world's most widely used linter.

SOFTWARE ENGINEERING & AI PROJECTS

PatentIQ | Autonomous AI Agent & RAG Engine | Python, Node.js, Fastify, LLMs, MCP, Pinecone, Prisma, PostgreSQL [GitHub](#)

- Architected an **autonomous AI agent** workflow using the **Model Context Protocol (MCP)** and dynamic tool calling, orchestrating vector retrieval tools, patent claim parsers, and external research endpoints.
- Engineered a low-latency **RAG** pipeline utilizing local **LLMs (Qwen2.5 via Ollama)** and Pinecone vector embeddings to automate unstructured data analysis, semantic claim overlap, and quantitative novelty risk scoring.
- Built a modular Fastify and PostgreSQL backend with Clean Architecture, dependency injection, and schema validation, providing structured JSON outputs for multi-step agentic execution loops and high API reliability.

DB Backup CLI (DBVault) | Distributed Disaster Recovery & Tooling | TypeScript, Node.js, Redis, BullMQ, AWS S3, Docker [GitHub](#)

- Architected an automated multi-engine backup/restore CLI published on npm and Docker Hub with **120 automated tests**, supporting PostgreSQL, MySQL, MongoDB, and SQLite with automated scheduling and checksum validation.
- Designed an asynchronous distributed job execution system using **Redis and BullMQ**, enforcing worker concurrency, rate-limits, execution guards, and Redis-based distributed locking to guarantee zero data loss.
- Implemented authenticated AES-256-GCM encryption with local key rotation, stream-based compression, and automated cloud offloading via modular **AWS S3** object storage adapters.

EXPERIENCE

Web Developer

Feb. 2025 – Dec. 2025

Students Special Group

Student Organization

- Engineered a full-stack **Project Management Portal** with React, Node.js, and MySQL to track faculty projects, tasks, and workflow progress across academic departments.
- Designed and implemented REST APIs and normalized database schemas for a **Faculty Reward Points Dashboard** to track real-time faculty contributions and evaluation metrics.
- Mentored **50+ junior developers** in JavaScript, backend architectural patterns, and production software engineering best practices.

ACHIEVEMENTS & CERTIFICATIONS

- 1200+ DSA Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, spanning arrays, dynamic programming, trees, and graph algorithms.
- IEEE DevSpark Hackathon Winner** – Secured **1st place among 120+ teams** in Full Stack Development by building **Plantera**, an AI-powered environmental monitoring application.
- BIT Hackathon Winner** – Secured **1st place among 200+ teams** by architecting a full-stack **Leave Management System** under rapid hackathon time constraints.
- Certifications:** AWS Cloud Practitioner Essentials, Oracle Cloud Infrastructure (OCI 2025 Certified Foundations Associate), and NPTEL IIT Madras Python – Elite + Topper (Top 5%, 95% score).